

Self-Supervised and Contrastive Representation Learning on STL-10

DATA 266 — HW6-5 | Final Report
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Abstract

This report compares three ResNet-18 representation-learning approaches on the STL-10 dataset: (1) supervised learning using only 10% of the labeled training split, (2) rotation-based self-supervised learning, and (3) SimCLR-style contrastive learning. All ResNet-18 models were trained from scratch without pretrained weights. The same 500 labeled images were used for supervised training and for linear evaluation of the self-supervised encoders. SimCLR achieved the strongest classification performance with 52.49% test accuracy, followed by supervised learning at 40.85% and rotation SSL at 31.06%.

1. Experimental Setup

STL-10 contains 5,000 labeled training images, 8,000 test images, and 100,000 unlabeled images across ten classes. A fixed 500-image labeled subset was selected with seed 5996 and reused across Parts A, B, and C for a fair comparison.

Setting	Value
Dataset	STL-10
Backbone	ResNet-18
Pretrained weights	None
Random seed	5996
Device	NVIDIA L4 GPU
Labeled subset	500 images
Test set	8,000 images
SimCLR unlabeled subset	20,000 images
SimCLR temperature	0.2

2. Part A — Supervised Learning with Limited Labels

The supervised baseline used a ResNet-18 with a ten-class output layer. The complete network was trained end-to-end using only the fixed 500-image labeled subset for 15 epochs. Random cropping and horizontal flipping were used during training, while the test set used deterministic normalization.

Metric	Result
Training images	500
Epochs	15
Optimizer	Adam
Learning rate	0.001
Weight decay	0.0001
Final test loss	2.5828

Final test accuracy	40.85%
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The training accuracy rose above 80%, while test accuracy peaked around the low-40% range. This gap indicates that the model learned useful class information but also overfit the small labeled subset.

3. Part B — Rotation Self-Supervised Learning

For rotation SSL, the encoder was trained on all 100,000 unlabeled images. Each image was randomly assigned one of four rotation labels: 0°, 90°, 180°, or 270°. A four-class rotation head was trained for 15 epochs. After pretraining, the rotation head was removed, the encoder was frozen, and a linear ten-class classifier was trained for 20 epochs using the same 500 labeled images as Part A.

Metric	Result
Unlabeled images	100,000
Rotation classes	4
Pretraining epochs	15
Final rotation loss	0.3848
Rotation-task accuracy	85.47%
Linear-evaluation epochs	20
Frozen encoder trainable parameters	0
STL-10 test accuracy	31.06%

The 85.47% result measures the auxiliary rotation-prediction task, not STL-10 object classification. The lower 31.06% linear-evaluation accuracy shows that learning orientation and spatial-layout cues did not transfer as effectively to object category separation as the other representations.

4. Part C — SimCLR-Style Contrastive Learning

SimCLR used a deterministic subset of 20,000 unlabeled images. Two independent augmented views were generated for each image using random resized crop, horizontal flip, color jitter, and grayscale conversion. The ResNet-18 encoder produced 512-dimensional features, followed by a 512-to-128 multilayer projection head. The NT-Xent loss used normalized projection vectors, cosine similarity, and temperature $\tau = 0.2$.

Metric	Result
Unlabeled images	20,000
Batch size	128
Augmentations	Crop, flip, color jitter, grayscale
Projection head	512 → 512 → 128
Temperature	0.2
Pretraining epochs	15
Initial contrastive loss	5.4820
Final contrastive loss	2.3574
Linear-evaluation epochs	20
Frozen encoder trainable parameters	0
STL-10 test accuracy	52.49%

The contrastive loss decreased steadily from 5.4820 to 2.3574, indicating that the encoder learned to bring augmented views of the same image closer together while separating views from other images. The resulting representation transferred best to the ten-class STL-10 classifier.

5. Accuracy Comparison

Model	Evaluation procedure	Test accuracy
Supervised ResNet-18	End-to-end training on 500 labeled images	40.85%
Rotation SSL + linear classifier	Frozen encoder; linear classifier on 500 labeled images	31.06%
SimCLR + linear classifier	Frozen encoder; linear classifier on 500 labeled images	52.49%

SimCLR was the best-performing method. Its accuracy was 11.64 percentage points higher than the supervised baseline and 21.43 percentage points higher than rotation SSL. This result supports the value of contrastive pretraining when labeled data are limited but unlabeled images are available.

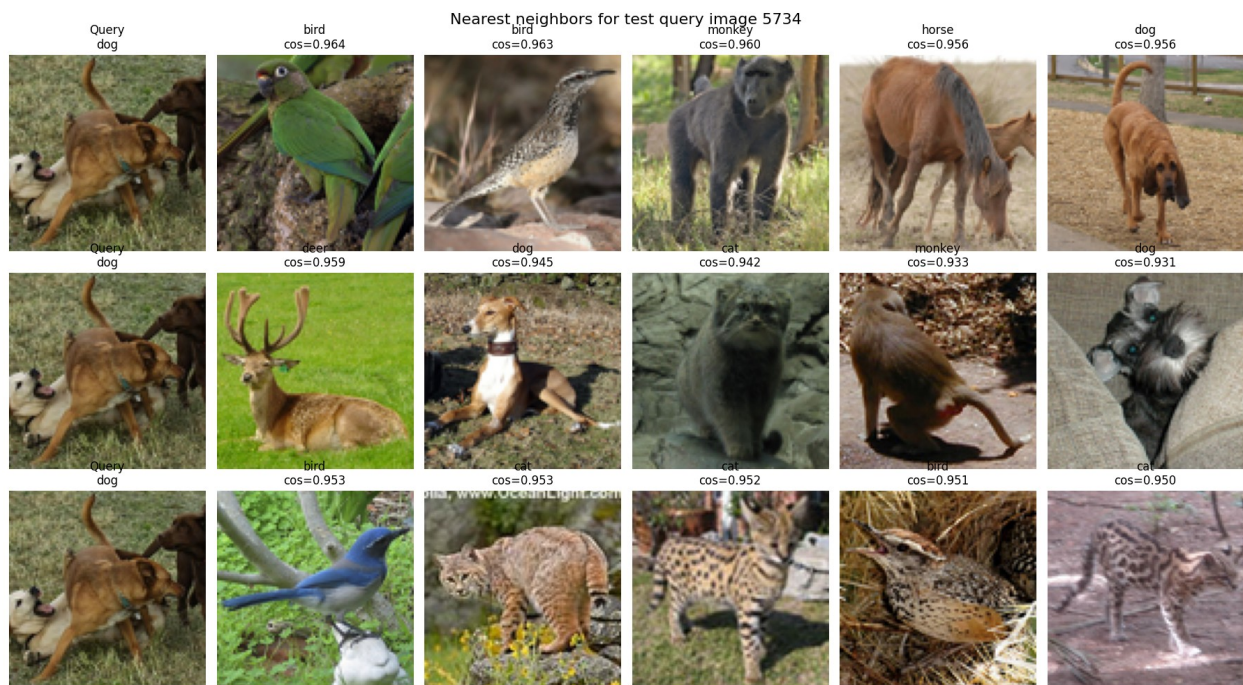
6. Nearest-Neighbor Representation Analysis

The test set was embedded with each encoder, producing 8,000 normalized 512-dimensional vectors per model. Three shared query images were used across all encoders: test index 5734 (dog), test index 61 (bird), and test index 1156 (monkey). For every query and encoder, the five nearest neighbors were retrieved using cosine similarity. The query itself was excluded from the neighbor list.

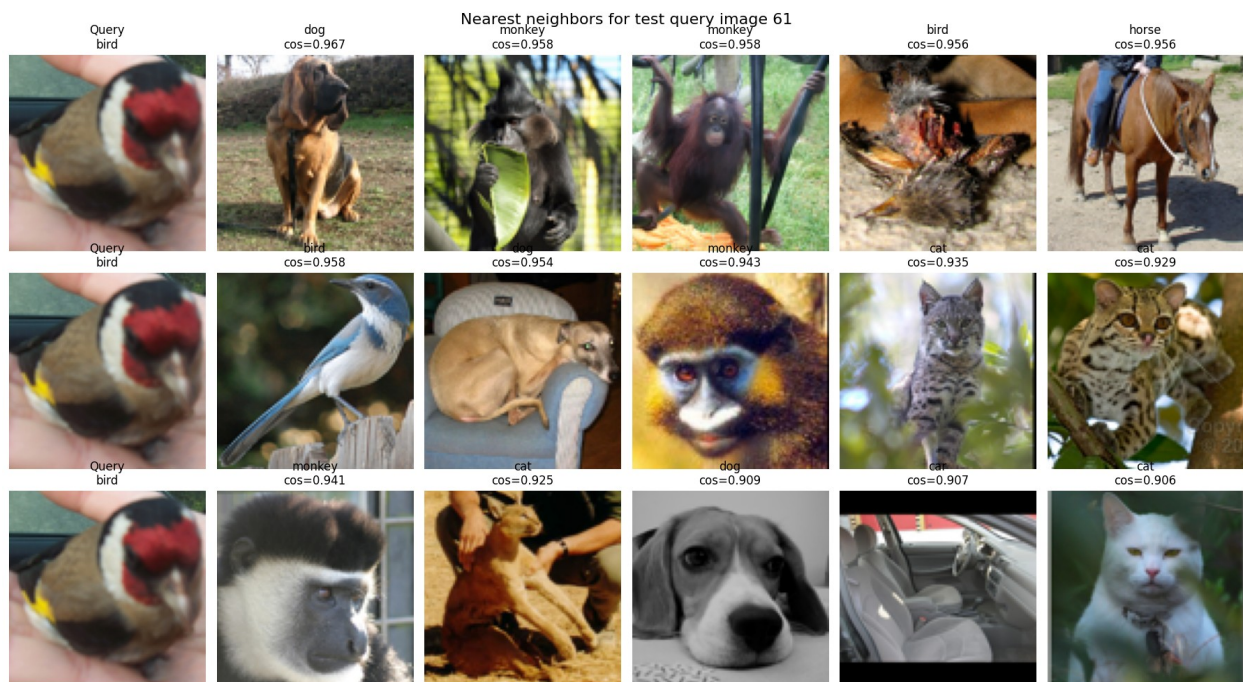
Query index	Class	Purpose
5734	Dog	Compare neighbor semantic similarity
61	Bird	Compare neighbor semantic similarity
1156	Monkey	Compare neighbor semantic similarity

The visualizations provide a qualitative view of representation quality. SimCLR is expected to produce the most semantically consistent neighbors because its contrastive objective encourages invariance to crop, flip, color, and grayscale changes. The supervised encoder can retrieve class-related images, but its features may be affected by overfitting to only 500 labeled examples. Rotation SSL learns useful edges, shapes, and spatial-layout cues, but its neighbors may share orientation or broad appearance without consistently belonging to the same object class. Confusions are especially plausible for visually similar categories such as cats and dogs or cars and trucks.

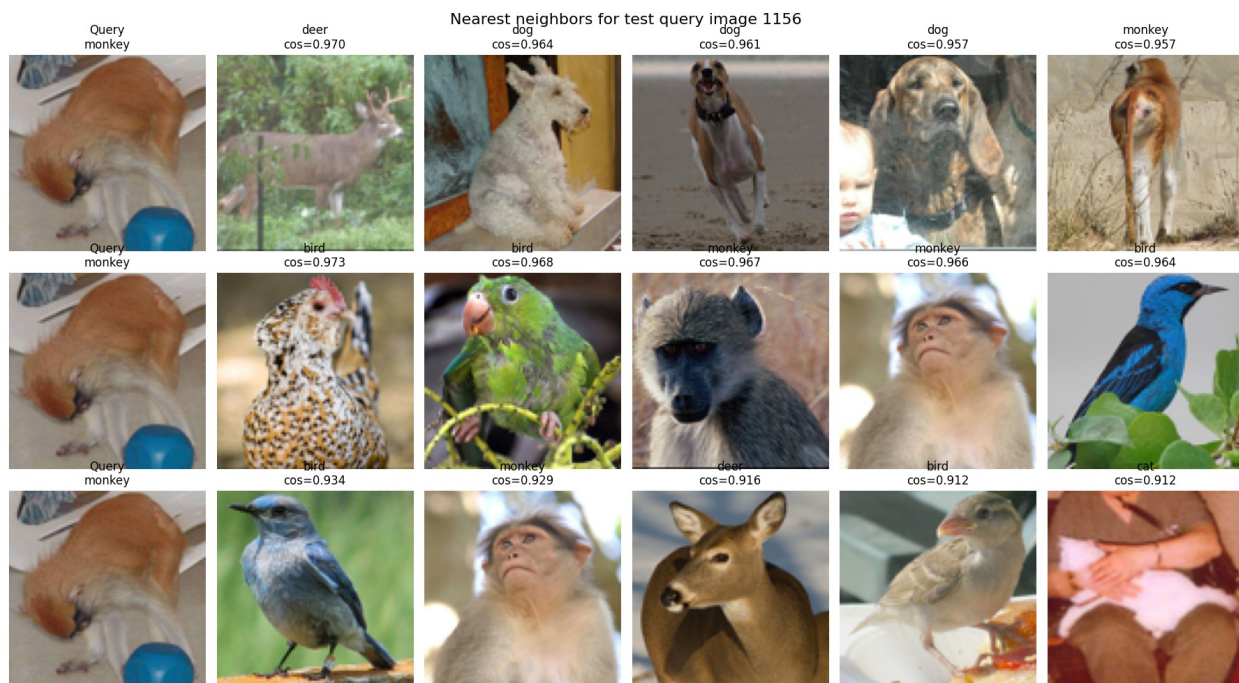
The following figures show the shared-query nearest-neighbor comparisons generated by the executed notebook.



Nearest-neighbor comparison figure 1



Nearest-neighbor comparison figure 2



Nearest-neighbor comparison figure 3

7. Improvement Suggestions

Supervised ResNet-18:

- Use stronger augmentation, dropout, and increased weight decay to reduce overfitting.
- Use a learning-rate scheduler and early stopping based on validation performance.
- Increase the amount of labeled data if available.

Rotation SSL:

- Combine rotation prediction with additional self-supervised tasks.
- Use more diverse geometric and color-based pretext tasks.
- Fine-tune some encoder layers during evaluation instead of keeping every layer frozen.

SimCLR:

- Use all 100,000 unlabeled images instead of the 20,000-image subset.
- Increase the batch size and train for more epochs.
- Use a cosine learning-rate schedule and tune the temperature parameter.
- Tune augmentation strength and projection-head size.

8. Reproducibility and Debugging

All experiments used seed 5996 for Python, NumPy, and PyTorch. The same fixed 500-image subset was reused across all three classification evaluations. The notebook was executed in Google Colab on an NVIDIA L4 GPU. During development, the initial linear-evaluation cell produced a NameError because it referenced DATA_DIR and labeled_indices, while the notebook defined DATA_ROOT and indices. The error was corrected, the cells were rerun, and the final reported results were produced after the correction.

9. AI Assistance Disclosure

AI assistance was used throughout this assignment to support understanding and organization of the requirements, planning the supervised, rotation SSL, and SimCLR experiments, writing and explaining portions of the PyTorch code, debugging undefined variables and dataset-path issues, implementing the contrastive loss and nearest-neighbor analysis, improving the training and evaluation workflow, creating plots and tables, interpreting the model results, developing improvement suggestions, checking the notebook against the rubric, and organizing the repository artifacts. The student executed the code in Google Colab, verified the outputs, reviewed the results, and made the final decisions about the implementation, analysis, and conclusions.

10. Conclusion

The experiments demonstrate that the learning objective strongly affects the quality of transferable visual representations. With only 500 labeled images available for classification, SimCLR produced the strongest representation and the highest test accuracy. Supervised learning performed reasonably well but showed signs of overfitting. Rotation SSL successfully learned the rotation task but produced features that transferred less effectively to STL-10 object recognition. The nearest-neighbor visualizations complement the accuracy results by showing how each encoder organizes images in embedding space.